

Introduction to Algorithm Design

HW03

Due date: 30 November 2015 Monday 23:59:59

Warning: You must write your algorithms in Python programming language. Otherwise, your work will not be graded.

Submission Policy: You can hand-in your solution to M. Rasih ELENLOLU. Alternatively, you can compress your solutions and submit from Moodle. Name of the compressed file must be exactly *sur-name_name1_name2_studentID.file_extension* where all of the letters must be lowercase and there should be no Turkish characters.

(e.g., Mahmud Rasih ÇELENLİOĞLU → celenlioglu_mahmud_rasih_123456789.zip)

- 1) Suppose that you have an array of mountain and city bikes in mixed order, e.g. M M C M C C C M, where M is a mountain bike and C is a city bike.). Size of the array is $2n$. Your goal is to put all mountain bikes at the beginning of the array. City bikes come after mount bikes. The only movement you are allowed make is to swap the position of two neighbor bikes. Design a brute force algorithm.
- 2) Suppose that you have a string. Your goal is to find the number of occurrences of a substring which starts with *firstChar* and ends with *lastChar*. For example, assume that you have a string TURKEY-TUNUSIAURUGUAY. Let *firstChar* be the letter T and *secondChar* be the letter Y. Number of occurrences of the substring is 3 (TURKEY, TURKEYTUNUSIAURUGUAY and TUNUSIAURUGUAY). Design a brute force algorithm that finds the number of occurrences of substrings.
- 3) Positional difference value (PDV) of any two strings with equal length is found by counting the positions at which corresponding letters are different. Design a brute force algorithm for the closest pair problem if the points in the problem are strings and the distance between any two points is determined by PDV.
- 4) Huseyin Kacmaz who died in 1990 at the age of 110 is one of our veteran soldiers who fought in the Dardanelles War. Suppose that his legacy will be partitioned among his son Turgut Kacmaz and his grand-daughter Eylul Kacmaz. The judge has a list of legacy items and their values in TL. The judge needs to partition the items in such a way that the total values of each side is equal. The judge is your mother and she needs you to make these complex calculations because you are studying Computer Engineering. Luckily, you are taking this class and will help your mother :) You decide to design a polynomial-time heuristic algorithm to partition items. Explain your algorithm using pseudocode and analyze the complexity of your algorithm.
- 5) (a) Design a DFS-based algorithm to check whether a graph is bipartite.
(b) Design a BFS-based algorithm to check whether a graph is bipartite.